

YARRAGUNTA CHARAN KUMAR REDDY

Embedded software

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PROFESSIONAL SUMMARY

B.Tech Electronics and Communication Engineering graduate with hands-on experience in Embedded C, C, C++, Python, STM32, Arduino, CAN, CAN FD, CAPL, UDS, ISO-TP, software testing, debugging, and automation. Strong foundation in electronics, instrumentation, microcontrollers, sensors, data acquisition concepts, and real-time embedded systems. Experienced in developing embedded applications, creating test cases, analyzing communication data, and automating software builds and tests using Jenkins. Seeking a Embedded Software Testing opportunity test and measurement, and industrial automation.

CORE SKILLS

- Programming: C, Embedded C, C++, Python
- Embedded Systems: STM32, Arduino, GPIO, ADC, PWM, Timers, Interrupts, Sensors, Actuators
- Communication Protocols: CAN, CAN FD, UART, SPI, I2C, Ethernet
- Automotive Software: CAPL, UDS, ISO-TP, ECU Diagnostics, CANoe, CANalyzer
- Testing and Automation: Software Testing, Debugging, Test Case Development, Unit Testing, Integration Testing, Jenkins, GoogleTest
- Tools and Development: CMake, Git Hub, VS Code, Jira
- Electronics and Instrumentation: Microcontrollers, Signal Monitoring, Sensor Data Acquisition, Measurement Fundamentals

PROJECTS

Battery Management System with Charge Monitoring and Fire Protection

- Designed an embedded battery monitoring and protection system using Arduino UNO, LM35 temperature sensor, ACS712 current sensor, and relay control.
- Implemented monitoring of State of Charge (SoC), State of Health (SoH), current, temperature, and charging status.
- Applied Coulomb Counting and Open Circuit Voltage (OCV) correction techniques for SoC estimation.
- Implemented CC-CV charging logic and safety monitoring for over-temperature and fire conditions.
- Used MATLAB/Simulink for system simulation and analysis; achieved approximately 95.4% SoC estimation accuracy.

CAN-Based Smart Key and Vehicle Simulator using STM32

- Developed an embedded vehicle simulator using STM32F407 Discovery and Controller Area Network (CAN) communication.
- Implemented GPIO control, CAN message transmission and reception, message ID handling, and real-time control logic.
- Integrated smart button start system functionality and analyzed communication between automotive system components.

Embedded Software Testing and Jenkins Automation

- Automated embedded software build and test execution using Jenkins.
- Developed test cases for vehicle functions including ignition modes and hazard light operation.
- Used C/C++, CMake, GoogleTest, and Jenkins for unit testing, integration testing, automated execution, and test result analysis.
- Debugged software failures by analyzing logs and test outputs.

EDUCATION

Bachelor of Technology (B.Tech), Electronics and Communication Engineering

Sri Venkateswara Institute of Technology, Anantapur | JNTUA | 2026 | CGPA: 7.5/10

Internship

Noveltronixs , Bengaluru | Trainee

TECHNICAL TRAINING & DOMAIN KNOWLEDGE.

- Embedded Systems and Embedded C
- Automotive CAN and CAN FD Communication
- Vector CANoe and CAPL
- STLC
- Jenkins and Software Test Automation